

Block Transfers, Safe Screening, and Linear Convergence for the Triangle Algorithm Family of SVM Solvers

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Abstract

The Triangle Algorithm (TA) solves hard-margin support vector machine training as a convex-hull distance problem with a certified duality gap. We extend the enhanced TA of Gupta and Kalantari to the full SVM family — L_2 and L_1 (hinge/ ν -SVM) soft margins and kernels — through exact reductions that reuse the algorithm’s dot-product caches, and we introduce three algorithmic improvements: *block weight transfers* that amortise each $O(n)$ scan over k MDM-style transfers via one exact line search along an aggregated direction; *gap-certified safe screening* that provably removes non-support points while preserving all certificates; and a *guarded step rule* with an away-step fallback. Our main theoretical result is global linear convergence of the guarded block algorithm with explicit constants and no combinatorial swap-step bounds, obtained by analysing the method on the Minkowski-difference polytope and inheriting the pyramidal-width machinery of Lacoste-Julien and Jaggi. Empirically, the optimised solver is the fastest method to a certified solution on six of seven benchmark classes against LIBSVM, LIBLINEAR, and SMO baselines, and every inequality in the analysis is verified numerically at machine precision.

1 Introduction

2 Background and setting

Let $V = \{v_1, \dots, v_n\}$, $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$ and

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} f(p, q) = \frac{1}{2} \|p - q\|^2. \quad (1)$$

With $z = p - q$ and the Minkowski-difference polytope $Z = \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$, (1) is the minimum-norm-point problem over Z , whose objective is 1-strongly convex and 1-smooth. The Triangle Algorithm [1, 2] maintains an iterate pair with convex weights (α, β) and a certified interval $[\text{LB}, \text{UB}]$ from parallel supporting hyperplanes; it stops when $\text{UB} - \text{LB} \leq \varepsilon \text{UB}$.

2.1 Exact reductions

L_2 soft margin: $K \mapsto K + I/C$ via slack coordinates $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$, never materialised (sparse diagonal corrections to the caches). ν -SVM: reduced convex hulls $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$, $\mu = 2/(\nu\ell)$; the extreme-point oracle becomes a capped top- k blend. Kernels: all quantities are inner products; iterates live in weight space.

*Draft skeleton generated from the replication and extension study; see `docs/block_convergence.md` and `REPORT.md` in the accompanying repository.

3 The guarded block-transfer algorithm

Algorithm 1 Guarded block transfer with away fallback (one side, one iteration)

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1: scores  $s_i \leftarrow \langle g, v_i \rangle$ ; top- $k$  receivers, worst- $k$  active donors, paired best-with-worst; pair 1 is
   the MDM pair  $(r^*, u^*)$ 
2: per-pair steps  $\gamma_j \leftarrow \min(\text{gap}_j/s_j, c_j)$ 
3: if pair 1 uncapped then
4:   take the better of the block step (exact line search along  $d = \sum_j \gamma_j d_j$ ) and the single
   MDM step ▷ guard
5: else
6:   if away step on  $u^*$  is boundary-clipped then
7:     take it ▷ drop step:  $\alpha_{u^*} \rightarrow 0$ 
8:   else
9:     take the best of {block, MDM, away( $u^*$ ), toward( $r^*$ )}
10:  end if
11: end if

```

4 Analysis

All proofs are in the repository note `docs/block_convergence.md` and will be inlined here.

Lemma 1 (Gap separability). *With the product representation $\lambda_{ij} = \alpha_i \beta_j$ over Z , $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$, and likewise for the Frank–Wolfe and away gaps; side transfers are pairwise directions of Z .*

Lemma 2 (Block gain). *With $S = \langle g, d \rangle$, $D = \|d\|^2$: $S \geq \sum_j \gamma_j^2 s_j$, $D \leq kS$, and the realised block gain satisfies $\Delta_B \geq S/(2k) \geq \Delta_{\text{MDM}}/(2k)$.*

Theorem 1 (Global linear convergence). *For any block size $k \geq 1$, the guarded block-transfer Triangle Algorithm with away fallback satisfies, with $\delta = \text{PW}(Z)$, $M = \text{diam}(Z)$, $\rho = c\delta^2/M^4$ for an explicit absolute constant c : every non-drop iteration contracts, $h_{t+1} \leq (1 - \rho)h_t$; drop iterations number at most $kT_{\text{good}} + s_0$; hence $h_T \leq h_0(1 - \rho)^{(T-s_0)/(k+2)}$.*

Proposition 1 (Near-orthogonal speedup). *If $|\langle d_i, d_j \rangle| \leq \eta\sqrt{s_i s_j}$ for $i \neq j$, the interior block gain is at least $(\sum_j \Delta_j)/(1 + \eta(k - 1))$.*

Theorem 2 (Safe screening). *With $r = \sqrt{\text{UB}^2 - \text{LB}^2}$, any zero-weight point with $(\langle h, v_i \rangle - \langle h, v_{\min} \rangle) > r\|v_i - v_{\min}\|$ is outside the optimal support; removing all such points preserves the optimum and the validity of all subsequent lower bounds.*

5 Experiments

6 Related work

7 Conclusion

References

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